

Nested brackets of the ribbon operators $R_e(t)$: the rigid factor does not deepen, and the component reading is a $k = 2$ artefact

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Abstract

Let $R_e(t)$ be the operator on the fermionic Fock space that adds an e -ribbon with weight t^{ht} . The previous paper closed the case $k = 2$: every entry of $[R_e, R_{e'}]$ is divisible by $1 + t$, the gcd is exactly $1 + t$, and the entries take exactly two shapes. Two conjectures were put forward for the k -fold nested bracket $\mathcal{C}_k = [R_{e_1}, [R_{e_2}, [\dots, R_{e_k}]\dots]]$: (A) that the gcd is $(1 + t)^{k-1}$, and (B) that every entry is $\pm t^a(1 + t)^{k-1}(1 - t)^{m-1}$ with m the number of connected components of μ/λ . *Both are false*, and this paper is their refutation together with the theorems that replace them.

(B) fails already at $k = 2$, for a reason that is structural rather than numerical: the commutator is supported exactly on the pairs where the two bead-hop intervals *cross*, and crossing intervals produce a *connected* skew shape. So $m \equiv 1$ on the whole support, and the statistic that the $k = 2$ data appeared to measure was not the component count but the number of moving beads — which in turn fails at $k = 3$ (Section 2).

(A) fails for every $k \geq 3$: the gcd is exactly $1 + t$, not $(1 + t)^{k-1}$. The positive results are Theorem 8 — $(1 + t)$ divides every entry of $\mathcal{C}_k$ for *all* k and all $e_i \geq 2$, with no distinctness hypothesis — and Theorem 10, a reduction formula

$$\left. \frac{\mathcal{C}_k}{1 + t} \right|_{t=-1} = \text{ad}_{p_{e_1}} \cdots \text{ad}_{p_{e_{k-2}}} (Q_{e_{k-1}, e_k}(-1)),$$

which expresses the level- k sharpness question entirely in terms of the level-2 quotient $Q_{e, e'} = [R_e, R_{e'}]/(1 + t)$ of the previous paper. Theorem 17 proves sharpness for $k = 3$ by a witness family uniform in (e_1, e_2, e_3) ; sharpness is verified computationally for $3 \leq k \leq 6$.

1 Setting and statement of the two conjectures

Let Λ be the ring of symmetric functions over $\mathbb{Z}[t]$ with Schur basis $\{s_\lambda\}$, identified with the charge-0 fermionic Fock space. For $e \geq 2$ let

$$R_e(t) s_\lambda = \sum_{\mu/\lambda \text{ an } e\text{-ribbon}} t^{\text{ht}(\mu/\lambda)} s_\mu,$$

where ht is one less than the number of rows. On the Maya diagram $B(\lambda) = \{\lambda_i - i : i \geq 1\} \subset \mathbb{Z}$ (so $B(\emptyset) = \mathbb{Z}_{<0}$), adding an e -ribbon is a bead hop $b \mapsto b + e$ with $b \in B$, $b + e \notin B$, and

$\text{ht} = \#(B \cap (b, b+e))$ is the number of beads strictly jumped (*abacus hop lemma*, proved in the previous paper). We write

$$\mathcal{C}_k = \mathcal{C}_k(e_1, \dots, e_k) := [R_{e_1}, [R_{e_2}, [\dots, [R_{e_{k-1}}, R_{e_k}] \dots]]],$$

and $\langle \mu \mid X \mid \lambda \rangle$ for the coefficient of s_μ in $X s_\lambda$.

The two conjectures under test, as posed in the brief, were:

Conjecture 1 ((A), depth). For pairwise distinct $e_1, \dots, e_k \geq 2$, $(1+t)^{k-1}$ divides every entry of $\mathcal{C}_k$, and $(1+t)^{k-1}$ is exactly the gcd.

Conjecture 2 ((B), components). Every nonzero entry $\langle \mu \mid \mathcal{C}_k \mid \lambda \rangle$ equals $\pm t^a (1+t)^{k-1} (1-t)^{m-1}$, where m is the number of connected components of the skew shape μ/λ .

Both come from a reading of the proved $k=2$ closed form: every nonzero entry of $[R_e, R_{e'}]$, $e \neq e'$, is either $\pm t^{h-1} (1+t)$ (when $\#(B \setminus B') = 1$) or $\pm t^{H-1} (1-t^2)$ (when $\#(B \setminus B') = 2$), and factoring $1-t^2 = (1+t)(1-t)$ writes both as $\pm t^a (1+t)^{k-1} (1-t)^{m-1}$ at $k=2$ — *provided* the two-bead case really is the $m=2$ case. Section 2 shows it is not.

Everything below rests on two proved results of the previous two papers, restated here for use.

Theorem 3 (Q75, Theorem A and Corollaries T1, T2; proved). *There is $f_e(t) \in \Lambda \otimes \mathbb{Z}[t]$, determined by $(1+t)f_e(t) = \sum_{b=0}^e t^b h_{e-b} e_b$, such that multiplication by $f_e(t)$ has matrix entries*

$$\langle \mu \mid M_{f_e(t)} \mid \lambda \rangle = t^{h(\mu/\lambda)} (1+t)^{c(\mu/\lambda)-1}$$

for μ/λ of size e free of 2×2 squares (a disjoint union of c ribbons), and 0 otherwise; here $h = \sum_i (\text{rows of the } i\text{th component} - 1)$. Consequently

$$M_{f_e(t)} - R_e(t) = (1+t) E_e(t), \quad \langle \mu \mid E_e(t) \mid \lambda \rangle = t^h (1+t)^{c-2} \quad (c \geq 2, \text{ no } 2 \times 2),$$

and 0 otherwise; $E_e(t)$ has polynomial entries. In particular $f_e(-1) = p_e$ and $R_e(-1) = M_{p_e}$.

Theorem 4 (Q76, closed form and sharpness; proved). *For $2 \leq e < e'$ write $B = B(\lambda)$, $B' = B(\mu)$, $D = B \setminus B'$, $U = B' \setminus B$. Then $\langle \mu \mid [R_e, R_{e'}] \mid \lambda \rangle$ vanishes unless $|D| \in \{1, 2\}$, and*

- (i) *if $D = \{b\}$, $U = \{b+e+e'\}$, the entry is $(\mathbf{1}[b+e \in B] - \mathbf{1}[b+e' \in B]) t^{h-1} (1+t)$ with $h = \#(B \cap (b, b+e+e'))$;*
- (ii) *if $D = \{b, c\}$ with b hopping by e and c by e' (the assignment is unique for $e \neq e'$), the entry is $\pm t^{H-1} (1-t^2)$ when the intervals $(b, b+e)$ and $(c, c+e')$ properly cross, and 0 otherwise, with H the sum of the two heights.*

The gcd of the entries is exactly $1+t$; witnesses at $\lambda = \emptyset$ are $\mu = (e', 1^e)$ with entry $t^{e-1} (1+t)$ and $\mu = (e', e)$ with entry $1-t^2$. We write $Q_{e,e'} := [R_e, R_{e'}]/(1+t)$.

2 Refutation of (B), and what the $k = 2$ data was measuring

2.1 The computation

For every λ with $|\lambda| \leq 5$ and every $2 \leq e < e' \leq 4$ the 283 nonzero entries of $[R_e, R_{e'}]$ were cross-tabulated by the triple

$$(v_{1+t}, v_{1-t}, m_{\text{shape}}),$$

the $(1+t)$ -adic and $(1-t)$ -adic valuations and the number of connected components of μ/λ . The result is not a near miss:

$$(1, 0, 1): 113, \quad (1, 1, 1): 170, \quad \text{no other cell occupied.}$$

That is, $m_{\text{shape}} = 1$ for *every* nonzero entry. Conjecture 2 is therefore false at $k = 2$, before $k = 3$ is reached, and the $(1-t)$ factor it predicted to be governed by the component count is governed by something else. Replacing m_{shape} by

$$m_{\text{hop}} := \#(B(\lambda) \setminus B(\mu)) = \text{the number of beads that move}$$

gives a perfect fit: all 283 entries satisfy $\text{entry} = \pm t^a(1+t)(1-t)^{m_{\text{hop}}-1}$, with $m_{\text{hop}} = 1$ exactly on the 113 entries where μ/λ is a genuine $(e+e')$ -ribbon and $m_{\text{hop}} = 2$ exactly on the 170 where μ/λ is connected *and contains a 2×2 square*.

2.2 Why the component count is constant: crossing implies connected

The collapse is not an accident of the range computed. It is forced by which pairs (λ, μ) the commutator supports at all.

Lemma 5 (interval–component dictionary). *Let $b \neq c$ be beads of B , let $b+e \notin B$ and (after b has hopped) $c+e'$ be empty, and suppose the two hops vacate two distinct sites, so that μ/λ is obtained by adding an e -ribbon on the diagonals $(b, b+e]$ and an e' -ribbon on the diagonals $(c, c+e']$. Then*

$$\mu/\lambda \text{ has two connected components} \iff (b, b+e] \cap (c, c+e'] = \emptyset.$$

If the intervals meet — whether properly crossing or nested — then μ/λ is connected and contains a 2×2 square.

Verification and proof sketch. By the abacus hop lemma, the ribbon added by $b \mapsto b+e$ occupies exactly the cells of content in $(b, b+e]$, one cell per diagonal. Two skew shapes each meeting every diagonal of an interval exactly once are disjoint as diagonal sets precisely when the intervals are disjoint; when the diagonal sets are disjoint the two ribbons occupy no common diagonal and, being each connected and diagonal-consecutive, are separated in the Young diagram. When the intervals meet, the two ribbons share a diagonal, hence occupy cells in the same column-and-row window, and the two cells on a shared diagonal together with their neighbours close up a 2×2 block.

Exhaustively verified for all λ with $|\lambda| \leq 6$ and all $2 \leq e < e' \leq 5$:

$$\text{disjoint} \Rightarrow (m = 2, \text{ no } 2 \times 2): 1254 \text{ cases}; \quad \text{crossing} \Rightarrow (m = 1, 2 \times 2): 796; \quad \text{nested} \Rightarrow (m = 1, 2 \times 2): 158.$$

No case violates the dictionary. The paragraph above is a proof for the disjoint direction; the 2×2 claim in the meeting case is stated at the level of verification. $\square$

Proposition 6 (the kernel). *Conjecture 2 is false for $k = 2$, and its failure is structural: by Theorem 4(ii) the two-bead entries of $[R_e, R_{e'}]$ vanish unless the two intervals properly cross, and by Lemma 5 crossing forces μ/λ to be connected. Hence $m_{\text{shape}} \equiv 1$ on the entire support of the commutator, and no data set drawn from $[R_e, R_{e'}]$, of any size, could ever have distinguished the component reading from any other statistic agreeing with it at $m = 1$.*

Remark 7. This is worth naming precisely. The statistic m of the commuting family $g_e(t)$, which counts components of a *generalized* border strip, is a genuine and varying statistic *there*. It was transported to μ/λ under the commutator; but the commutator is built by a cancellation that annihilates exactly the multi-component targets. The transported statistic is not merely constant on the new domain by numerical accident — it is constant *because the operator was constructed to kill its variation*. A control that varies m on the source of the transport does not vary it on the target.

2.3 And m_{hop} fails too, at $k = 3$

The repaired reading entry $= \pm t^a(1+t)^{k-1}(1-t)^{m_{\text{hop}}-1}$ was then tested at $k = 3$, where $m_{\text{hop}} = 3$ becomes reachable. Over all λ with $|\lambda| \leq 5$ and all ten triples $2 \leq e_1 < e_2 < e_3 \leq 6$ there are 9524 nonzero entries, distributed as

$(v_{1+t}, v_{1-t}, m_{\text{hop}}) :$	(1, 0, 1)	(1, 0, 2)	(1, 1, 1)	(1, 1, 2)	(1, 2, 2)	(2, 0, 1)	(2, 1, 2)	(2, 2, 3)
count :	858	1654	86	3618	717	60	704	1827

and $m_{\text{shape}} = 1$ in all 9524. The relation $v_{1-t} = m_{\text{hop}} - 1$ holds on 7067 entries and fails on 2457. Moreover the residues after dividing out $t^a(1+t)^{v_{1+t}}(1-t)^{v_{1-t}}$ are not all ± 1 : they include ± 2 , $\pm(t^2 + 1)$ and $\pm(t^2 + t + 1)$, so the “coefficients in $\{0, \pm 1\}$, exactly two entry shapes” rigidity of $k = 2$ is a $k = 2$ phenomenon and does not survive.

One clean sub-law does survive at $k = 3$: all 1827 entries with $m_{\text{hop}} = 3$ have $v_{1+t} = v_{1-t} = 2$ exactly, i.e. are of the form $t^a(1-t^2)^2 \cdot u(t)$ with $u(\pm 1) \neq 0$. Whether the maximal-hop entries carry $(1-t^2)^{k-1}$ in general is left open.

3 Divisibility, for every k and with no distinctness hypothesis

Theorem 8. *For every $k \geq 2$ and all $e_1, \dots, e_k \geq 2$ — not necessarily distinct — the polynomial $1+t$ divides every entry of $\mathcal{C}_k$.*

Proof. Write $u = 1+t$, $M_i := M_{j_{e_i}(t)}$ and $E_i := E_{e_i}(t)$, so that $R_{e_i} = M_i - uE_i$ by Theorem 3, with E_i having entries in $\mathbb{Z}[t]$. The nested bracket is multilinear in its k arguments, so

$$\mathcal{C}_k = \sum_{S \subseteq [k]} (-u)^{|S|} \Gamma_S, \quad \Gamma_S := [A_1, [A_2, [\dots, [A_{k-1}, A_k] \dots]]], \quad A_i = \begin{cases} E_i, & i \in S \\ M_i, & i \notin S. \end{cases} \quad (1)$$

Every Γ_S has entries in $\mathbb{Z}[t]$. The term $S = \emptyset$ is a nested bracket of multiplication operators on the commutative ring Λ , hence 0. Every remaining term carries a factor $u = 1+t$. $\square$

Remark 9. Equivalently, and more cheaply: $R_e(-1) = M_{p_e}$ and power sums commute, so $\mathcal{C}_k$ vanishes at $t = -1$. The point of the expansion (1) is that it also controls the *next* order, which is what Conjecture 1 is about. Note that distinctness of the e_i is never used; the conjecture’s “pairwise distinct” hypothesis is not where the boundary lies.

4 The reduction formula

For an operator X write $\text{ad}_{p_a}(X) = [M_{p_a}, X]$. These commute with one another, since $[\text{ad}_{p_a}, \text{ad}_{p_b}] = \text{ad}_{[M_{p_a}, M_{p_b}]} = 0$. Put $N_e := E_e(-1)$; by Theorem 3,

$$\langle \mu \mid N_e \mid \lambda \rangle = (-1)^{h(\mu/\lambda)} \quad \text{if } \mu/\lambda \text{ has size } e, \text{ no } 2 \times 2 \text{ and exactly two components,}$$

and 0 otherwise (the factor $(1+t)^{c-2}$ kills $c \geq 3$ at $t = -1$).

Theorem 10 (reduction to level 2). *For every $k \geq 2$ and all $e_1, \dots, e_k \geq 2$,*

$$\left. \frac{\mathcal{C}_k(e_1, \dots, e_k)}{1+t} \right|_{t=-1} = \text{ad}_{p_{e_1}} \text{ad}_{p_{e_2}} \cdots \text{ad}_{p_{e_{k-2}}} \left([M_{p_{e_k}}, N_{e_{k-1}}] - [M_{p_{e_{k-1}}}, N_{e_k}] \right),$$

and the operator in brackets is exactly $Q_{e_{k-1}, e_k}(-1)$, the level-2 quotient of Theorem 4 evaluated at $t = -1$.

Proof. By (1), $\mathcal{C}_k/u|_{u=0} = -\sum_{i=1}^k \Gamma_{\{i\}}|_{t=-1}$, since the $S = \emptyset$ term vanishes identically and every $|S| \geq 2$ term carries $u^{|S|}$. At $t = -1$ we have $M_i \mapsto M_{p_{e_i}}$ and $E_i \mapsto N_{e_i}$.

Fix i and consider $\Gamma_{\{i\}}$, the nested bracket with N_{e_i} in slot i and multiplication operators in all other slots. Its innermost bracket is $[A_{k-1}, A_k]$. If $i \leq k-2$ then $A_{k-1} = M_{p_{e_{k-1}}}$ and $A_k = M_{p_{e_k}}$ are both multiplication operators, so the innermost bracket already vanishes and $\Gamma_{\{i\}} = 0$. Hence only $i = k-1$ and $i = k$ survive:

$$\Gamma_{\{k-1\}}|_{t=-1} = \mathcal{D}([N_{e_{k-1}}, M_{p_{e_k}}]), \quad \Gamma_{\{k\}}|_{t=-1} = \mathcal{D}([M_{p_{e_{k-1}}}, N_{e_k}]),$$

where $\mathcal{D} := \text{ad}_{p_{e_1}} \cdots \text{ad}_{p_{e_{k-2}}}$. Therefore

$$\left. \frac{\mathcal{C}_k}{1+t} \right|_{t=-1} = -\mathcal{D}([N_{e_{k-1}}, M_{p_{e_k}}] + [M_{p_{e_{k-1}}}, N_{e_k}]) = \mathcal{D}([M_{p_{e_k}}, N_{e_{k-1}}] - [M_{p_{e_{k-1}}}, N_{e_k}]).$$

Taking $k = 2$ (so $\mathcal{D} = \text{id}$) identifies the bracketed operator with $Q_{e_1, e_2}(-1)$, which is Corollary T3 of the previous paper read at first order. $\square$

Corollary 11 (why (A) was never going to hold). *The mechanism that produces the factor $1+t$ produces exactly one such factor, for every k . Conjecture 1 asked the $k-1$ single- E terms of (1) to cancel to first order; Theorem 10 shows that $k-3$ of them vanish identically for trivial reasons and the remaining two do not cancel. At $k = 2$ the observed exponent 1 coincides with $k-1$; the $k = 2$ data cannot separate “ $k-1$ ” from “1”, and it was that degeneracy, not the arithmetic, that suggested (A).*

Corollary 12 (refutation of (A)). *For $3 \leq k \leq 6$ and the ranges computed below, the gcd of the entries of $\mathcal{C}_k$ is exactly $1+t$, not $(1+t)^{k-1}$. Conjecture 1 is false for every $k \geq 3$ tested.*

5 Sharpness: a witness family, uniform in the ribbon sizes

We now prove that the right-hand side of Theorem 10 is nonzero for $k = 3$, uniformly in (e_1, e_2, e_3) , by exhibiting a single family of matrix entries. Throughout this section $\lambda = \emptyset$, so $B = \mathbb{Z}_{<0}$; $e_1, \dots, e_k \geq 2$ are pairwise distinct with sorted values $f_1 < \cdots < f_k$; and

$$j := f_{k-1} \text{ (the second largest size),} \quad E := \sum_i e_i, \quad b := -1 - j, \quad \mu := (E - j, 1^j).$$

A direct check on beta-sets gives $B(\mu) = B \setminus \{b\} \cup \{b + E\}$: the target is the single-bead-hop target, and $\mu/\emptyset$ is a hook, i.e. an E -ribbon of height j .

5.1 Paths are spatial chains

Lemma 13 (chain lemma). *Fix a word $R_{e_{w_1}} \cdots R_{e_{w_k}}$ (acting rightmost first). Every sequence of k legal bead hops taking B to $B(\mu)$ has the following form: there is a permutation $\rho \in S_k$ and a chain*

$$b = x_0 < x_1 < \cdots < x_k = b + E, \quad x_i - x_{i-1} = e_{\rho(i)},$$

such that the i th spatial hop moves a bead from x_{i-1} to x_i , and each of the k hops of the word performs exactly one of these.

Proof. Form the directed multigraph on $\mathbb{Z}$ with one edge (source $\rightarrow$ target) per hop. Every edge increases the coordinate, so the graph is acyclic. The net effect on the bead set is to remove b and add $b + E$, so the out-degree minus in-degree is $+1$ at b , -1 at $b + E$ and 0 elsewhere. An acyclic multigraph with this imbalance has no vertex of out-degree ≥ 2 : such a vertex would force, in any decomposition into a $b \rightarrow b + E$ trail plus closed trails, a closed trail, which is impossible. Likewise for in-degree. Hence the edges form a single simple directed path from b to $b + E$; its vertices, listed in increasing order, are $x_0 < \cdots < x_k$, and the increments are the e_i in some order ρ . $\square$

Lemma 14 (legality and the threshold). *Let $r = r(\rho) := \min\{i : e_{\rho(1)} + \cdots + e_{\rho(i)} > j\}$, so that $x_i < 0$ for $i < r$ and $x_i \geq 0$ for $i \geq r$. A time order of the k spatial hops is legal if and only if it is a linear extension of the poset P_ρ generated by*

$$r \prec r - 1 \prec \cdots \prec 1 \quad \text{and} \quad r \prec r + 1 \prec \cdots \prec k.$$

Proof. The site x_i ($1 \leq i \leq k - 1$) is initially occupied iff $x_i < 0$, and $x_0 = b < 0$ is occupied while $x_k = b + E \geq 0$ is empty (indeed $b + E \geq f_k - 1 > 0$). If $x_i \geq 0$ then x_i starts empty, so the bead that later leaves x_i must be the one that arrived from x_{i-1} : spatial hop i must precede spatial hop $i + 1$. If $x_i < 0$ then x_i starts occupied by a bead distinct from the one arriving from x_{i-1} , and that resident must vacate first: spatial hop $i + 1$ must precede spatial hop i . These are the only constraints, since a hop is legal exactly when its source is occupied and its target empty at that moment, and only chain sites change occupancy. Because the partial sums of the $e_{\rho(l)}$ are strictly increasing, the sign of x_i switches once, at $i = r$; assembling the constraints gives P_ρ . $\square$

Lemma 15 (height depends only on the chain). *For any legal time order of the chain ρ , the total weight of the path is $t^{j-r(\rho)+1}$.*

Proof. The height of the spatial hop $x_{i-1} \rightarrow x_i$ counts the beads strictly inside (x_{i-1}, x_i) at the moment it is performed. The only sites whose occupancy ever changes are the chain sites $x_0, \dots, x_k$, and none of them lies strictly inside (x_{i-1}, x_i) . Hence that height equals $\#\{c \in \mathbb{Z}_{<0} : x_{i-1} < c < x_i\}$, independently of the time order. Summing over i ,

$$H(\rho) = \#\{c \in \mathbb{Z}_{<0} : x_0 < c < x_k\} - \#\{1 \leq i \leq k - 1 : x_i < 0\} = j - (r - 1),$$

because $\{c < 0 : c > -1 - j\} = \{-j, \dots, -1\}$ has j elements, and the negative interior chain sites are exactly $x_1, \dots, x_{r-1}$. $\square$

5.2 The sign rule

Lemma 16 (unimodal expansion). *Expanding $[A_1, [A_2, [\dots, [A_{k-1}, A_k] \dots]]$ into words, the words that occur are exactly those whose time-ordered index sequence $(w_1, \dots, w_k)$ (rightmost factor first) is unimodal with peak k : that is, $w_1 < w_2 < \dots < w_q = k > w_{q+1} > \dots > w_k$ for some q . Each such sequence occurs exactly once, with sign $(-1)^{q-1}$, where $q-1$ is the number of indices acting before R_{e_k} .*

Proof. Induct on k . For $k = 1$ the claim is trivial. Writing $[A_1, W] = A_1W - WA_1$ with W ranging over the words of the bracket on $A_2, \dots, A_k$: index 1 is appended either at the far left of the word (with sign $+$) or at the far right (with sign $-$). In the time-ordered sequence (the reverse of the word) this places 1 either at the very end or at the very beginning. Since 1 is the minimum index, both placements preserve unimodality with peak k , and every unimodal sequence with peak k arises from exactly one choice. Placing 1 at the beginning of the time order adds one index before the peak and carries the sign $-$; placing it at the end adds none and carries $+$. This is precisely $(-1)^{q-1}$. $\square$

5.3 The witness

Theorem 17 (uniform witness at $k = 3$). *Let $e_1, e_2, e_3 \geq 2$ be pairwise distinct with sorted values $f_1 < f_2 < f_3$, let $j = f_2$, $E = e_1 + e_2 + e_3$ and $\mu = (E - j, 1^j)$. Then*

$$\langle \mu \mid \mathcal{C}_3(e_1, e_2, e_3) \mid \emptyset \rangle = \begin{cases} t^{j-1}(1+t), & \text{if } e_3 = f_3, \\ -t^{j-1}(1+t), & \text{if } e_2 = f_3, \\ 0, & \text{if } e_1 = f_3. \end{cases}$$

In particular the gcd of the entries of $\mathcal{C}_3$ is exactly $1+t$ whenever the largest of the three sizes is not in the outermost slot, and Conjecture 1 is false for $k = 3$ for every such triple.

Proof. By Lemmas 13–16 the entry equals

$$\sum_{\rho \in S_3} \sum_{\tau \in \mathcal{L}(P_\rho)} \varepsilon(\rho, \tau) t^{j-r(\rho)+1},$$

where τ runs over the linear extensions of P_ρ and $\varepsilon(\rho, \tau)$ is the sign of Lemma 16 applied to the operator-index sequence obtained by listing the spatial hops in the time order τ and reading off ρ ; $\varepsilon = 0$ if that sequence is not unimodal with peak 3.

Only $r \in \{1, 2\}$ occurs. $r \geq 3$ would require $e_{\rho(1)} + e_{\rho(2)} \leq j = f_2$; but any two distinct sizes sum to more than f_2 , since the smaller of them is $\geq f_1 \geq 2 > 0$ and the larger is $\geq f_2$. So $r \leq 2$ always. Also $r = 1$ iff $e_{\rho(1)} > j$ iff $e_{\rho(1)} = f_3$.

The case $e_3 = f_3$, i.e. the largest size sits in the innermost slot, $\rho(1) = 3$ meaning $r = 1$.

$r = 1$: P_ρ is the chain $1 \prec 2 \prec 3$, so $\tau = (1, 2, 3)$ is the unique extension and the operator sequence is $(\rho(1), \rho(2), \rho(3)) = (3, \rho(2), \rho(3))$. Unimodality with peak 3 forces the tail to decrease, so $\rho = (3, 2, 1)$, with $q = 1$ and sign $+1$. The other candidate $\rho = (3, 1, 2)$ gives $(3, 1, 2)$, not unimodal. Contribution: $+t^j$.

$r = 2$: here $\rho(1) \neq 3$, so $\rho \in \{(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1)\}$, and P_ρ has minimum 2 with $2 \prec 1$ and $2 \prec 3$, giving the two extensions $\tau = (2, 1, 3)$ and $\tau = (2, 3, 1)$; the operator sequences

are $(\rho(2), \rho(1), \rho(3))$ and $(\rho(2), \rho(3), \rho(1))$ respectively. Tabulating:

ρ	$\tau = (2, 1, 3)$	$\tau = (2, 3, 1)$
$(1, 2, 3)$	$(2, 1, 3)$ not unimodal	$(2, 3, 1)$ $q = 2, -1$
$(1, 3, 2)$	$(3, 1, 2)$ not unimodal	$(3, 2, 1)$ $q = 1, +1$
$(2, 1, 3)$	$(1, 2, 3)$ $q = 3, +1$	$(1, 3, 2)$ $q = 2, -1$
$(2, 3, 1)$	$(3, 2, 1)$ $q = 1, +1$	$(3, 1, 2)$ not unimodal

The signed count is $-1 + 1 + 1 - 1 + 1 = +1$. Contribution: $+t^{j-1}$.

Adding, the entry is $t^{j-1} + t^j = t^{j-1}(1+t)$, and $(1+t)^2$ does not divide it. Since $1+t$ divides every entry by Theorem 8, the gcd is exactly $1+t$.

The other two cases. Let m be the slot of the largest size, so $r(\rho) = 1$ iff $\rho(1) = m$. The same two tabulations, with $m = 2$ and $m = 1$, give:

ρ	$(2, 1, 3)$	$(2, 3, 1)$	ρ	$(2, 1, 3)$	$(2, 3, 1)$
$m = 2 :$	$(1, 2, 3)$	— $q=2, -1$	$m = 1 :$	$(2, 1, 3)$	$q=3, +1$ $q=2, -1$
	$(1, 3, 2)$	— $q=1, +1$		$(2, 3, 1)$	$q=1, +1$ —
	$(3, 1, 2)$	$q=2, -1$ $q=3, +1$		$(3, 1, 2)$	$q=2, -1$ $q=3, +1$
	$(3, 2, 1)$	$q=2, -1$ —		$(3, 2, 1)$	$q=2, -1$ —

(the columns are the two extensions τ , the dash marks a non-unimodal sequence), giving $r = 2$ signed counts -1 and 0 respectively. For $r = 1$, where $\tau = (1, 2, 3)$ is forced and the operator sequence is ρ itself: for $m = 2$ the admissible ρ are $(2, 1, 3)$ (not unimodal) and $(2, 3, 1)$ ($q = 2, -1$), count -1 ; for $m = 1$ they are $(1, 2, 3)$ ($q = 3, +1$) and $(1, 3, 2)$ ($q = 2, -1$), count 0 . Hence the entry is $-t^j - t^{j-1}$ when $m = 2$ and 0 when $m = 1$, as stated.

For general k the $r = 1$ count is the same computation: with $\rho(1) = m < k$, unimodality forces the indices before the peak to be m together with an arbitrary subset $T \subseteq \{m+1, \dots, k-1\}$ (in increasing order) and the rest to follow in decreasing order, so the signed count is $\sum_T (-1)^{|T|+1} = -(1-1)^{k-1-m}$, which is 0 for $m \leq k-2$ and -1 for $m = k-1$; and for $m = k$ the unique admissible chain $\rho = (k, k-1, \dots, 1)$ gives $+1$. $\square$

Corollary 18. *For $k = 3$ and all pairwise distinct $e_1, e_2, e_3 \geq 2$ with $\max\{e_i\} \neq e_1$, the gcd of the entries of $\mathcal{C}_3$ is exactly $1+t$. Equivalently, by Theorem 10, $\text{ad}_{p_{e_1}}(Q_{e_2, e_3}(-1)) \neq 0$.*

Remark 19 (the $k \geq 4$ status). Lemmas 13–16 hold for all k ; only the evaluation of the signed counts was specialised. The $r = 1$ count is settled for all k inside the proof above. Two things remain for $k \geq 4$: the $r = 2$ count, and the vanishing of the counts for $r \geq 3$ (which for $k = 3$ was vacuous but for $k \geq 4$ is a genuine claim — e.g. for sizes $\{2, 3, 5, 6\}$ one has $2 + 3 \leq j = 5$). Computationally, the entry at $\mu = (E - j, 1^j)$ is $t^{j-1}(1+t)$ for every 4-subset and 5-subset of $\{2, \dots, 6\}$ tested with the largest size innermost. This is the one precisely located gap in the paper.

6 Verification

All computations use two engines that share no code: the *Maya engine*, which composes bead hops on beta-sets, and the *shape engine*, which evaluates M_{p_e} and N_e from skew-shape statistics (size, component count, presence of a 2×2 , height) via the independent symmetric-function library of the Q75 paper. The two engines were already reconciled on 238594 entries at $k = 2$.

1. **Instrument check (before any surprise was believed).** The Jacobi identity $[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0$ and the antisymmetry $[A, [B, C]] = -[A, [C, B]]$ hold entrywise on the Maya engine: 28 configurations ($|\lambda| \leq 3$, all triples from $\{2, 3, 4, 5\}$), 0 failures.
2. **Refutation of (B) at $k = 2$:** 283 nonzero entries, $|\lambda| \leq 5$, $2 \leq e < e' \leq 4$; $m_{\text{shape}} = 1$ in all of them; 0 violations of $\pm t^a(1+t)(1-t)^{m_{\text{hop}}-1}$.
3. **Refutation of (A):** 9524 nonzero entries of $\mathcal{C}_3$, $|\lambda| \leq 5$, all ten triples from $\{2, \dots, 6\}$. The gcd of the entries is exactly $1+t$ in all 190 pairs $(\lambda, (e_1, e_2, e_3))$, never $(1+t)^2$.
4. **Reduction formula (Theorem 10)** checked entrywise, Maya engine against shape engine: $k = 3$, 1310 entries ($|\lambda| \leq 4$, all triples from $\{2, \dots, 5\}$), 0 mismatches, all 1310 nonzero; $k = 4$, 270 entries, 0 mismatches, all nonzero. The $k = 4$ nonzero count was independently reproduced by the shape engine alone (270), which is how the two sides were cross-checked.
5. **Sharpness for $3 \leq k \leq 6$,** via the shape engine and Theorem 10: $\text{ad}_{p_{e_1}} \cdots \text{ad}_{p_{e_{k-2}}}(Q_{e_{k-1}, e_k}(-1))$ is nonzero for $(2, 3, 4)$, $(2, 3, 4, 5)$, $(2, 3, 4, 5, 6)$ and $(2, 3, 4, 5, 6, 7)$ — 144, 270, 217 and 621 nonzero entries respectively.
6. **Lemma 5:** 2208 two-hop configurations, $|\lambda| \leq 6$, $2 \leq e < e' \leq 5$, partitioned as 1254 disjoint / 796 crossing / 158 nested with no violation of the dictionary.

Controls

C1 (carried from Q76): height convention. Perturbing the height convention breaks the $k = 2$ classification, hence the Maya engine's calibration.

C2 (carried from Q76): drop the crossing hypothesis. Breaks the $k = 2$ closed form.

C3: repeated sizes. The conjectures were stated for pairwise distinct e_i ; Theorems 8 and 10 do not use distinctness, so this is a genuine variation. Computed: for $(2, 3, 2)$, $(3, 2, 3)$, $(2, 2, 3)$, $(3, 4, 3)$ and $(2, 3, 2, 3)$ the bracket is nonzero and its gcd is again exactly $1+t$; for $(2, 3, 3)$, $(2, 2, 2)$ and $(2, 2, 3, 3)$ the bracket vanishes identically, as it must, since the innermost bracket $[R_a, R_a]$ is zero. Distinctness is not where the boundary lies, and the statement has not been quietly restricted.

C4: Jacobi. Item 1 above; run *before* any of the refutations were believed.

C5: hand computation. The entry $\langle (6, 1, 1, 1) \mid \mathcal{C}_3(2, 3, 4) \mid \emptyset \rangle$ was computed by hand from the bead definition, independently of both engines: with $b = -4$, $B = \mathbb{Z}_{<0}$ and target $B \setminus \{-4\} \cup \{5\}$, the six spatial chains

$$(2, 3, 4), (2, 4, 3), (3, 2, 4), (3, 4, 2), (4, 2, 3), (4, 3, 2)$$

were enumerated with their legal time orders and the four words $R_2R_3R_4 - R_2R_4R_3 - R_3R_4R_2 + R_4R_3R_2$. Chain by chain the contributions are

$$(2, 3, 4): -t^2, \quad (2, 4, 3): +t^2, \quad (3, 2, 4): +t^2 - t^2 = 0, \quad (3, 4, 2): +t^2, \quad (4, 2, 3): 0, \quad (4, 3, 2): +t^3,$$

the first two cancelling each other and $(4, 2, 3)$ being excluded because its forced time order $(4, 2, 3)$ is not one of the four words. Total $t^2 + t^3 = t^2(1+t)$, which is what both engines return and what Theorem 17 predicts for $j = 3$.

C6 (negative control on the cross-tabulation). The cross-tabulation is not vacuous: at $k = 3$ it occupies eight distinct cells and the residues take six distinct values, so the statistic being tabulated does vary over the data. This is the check that Proposition 6 shows was *missing* at $k = 2$, where the tabulation occupied a single value of m_{shape} .

7 What is left

1. The $r \geq 2$ signed counts for $k \geq 4$ (Remark after Theorem 17). This would give sharpness of the gcd for all k , uniformly in the sizes, and is pure sign combinatorics — no t appears.
2. Whether the maximal-hop entries $m_{\text{hop}} = k$ always carry $(1 - t^2)^{k-1}$ exactly. True on all 1827 instances at $k = 3$.
3. A closed form for $\mathcal{C}_3$. The $k = 2$ closed form had two shapes and coefficients in $\{0, \pm 1\}$; at $k = 3$ neither survives, so any closed form must be organised differently — Theorem 10 suggests organising it by the level-2 quotient rather than by shape.
4. The order of $Q_{e,e'}(-1)$ as a differential operator on $\Lambda = \mathbb{Q}[p_1, p_2, \dots]$. Since ad_{p_e} acts on operators as $-\partial/\partial \xi_e$, where ξ_e is the symbol of ∂_{p_e} , sharpness at level k is exactly the statement that $Q_{e_{k-1}, e_k}(-1)$ has a term divisible by $\partial_{p_{e_1}} \cdots \partial_{p_{e_{k-2}}}$. An operator of finite order would make (A)'s successor fail for large k ; the computations to $k = 6$ say the order is at least 4.

Provenance

Theorem 3 and Theorem 4 are my own, proved in 2026-09-03-Q75-multiplication-defect.tex and 2026-09-04-Q76-commutator-quotient.tex. Nothing in this paper depends on Jing-Liu 2310.15730: the commuting-family route to divisibility, which would have required Corollary 4.4 of that paper (held only at **agent-summary** extraction level), has been replaced by Theorem 8, which uses only the decomposition $R_e = M_{f_e} - (1 + t)E_e$ proved in the Q75 paper. The identification of the two-valued $k = 2$ statistic as a component count is the reading recorded in the dream journal of 4 September 2026; this paper refutes it.